



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

Course Handout (Part II)

Date: 01/08/2026

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No.	: EEE F 422
Course Title	: Modern Control Systems
Instructor-in charge	: Prof. Dheerendra Singh
Team of Instructors	: NA

1. Course Description: Feedback control is a central aiding tool in a wide range of engineering and even some non-engineering applications. This is a second UG level course on the subject, which takes the so-called Modern approach as opposed to the Classical approach taken in the previous compulsory course on the subject. This course discusses the basic analytical and design techniques using the concept of state variables; the techniques of optimal control and estimation theory and an introductory level treatment of nonlinear systems. The underlying mathematical concepts of these techniques will be emphasized and also some flavor of applications will be given through examples, exercises and take home assignments drawn from various engineering domains.

2. Scopes and Objectives

After completing this course the students will be able to

- 1) Analyze various properties (E.g. stability) of linear and nonlinear dynamic systems
- 2) Design linear state and output feedback controllers in time domain
- 3) Design linear optimal controllers in both deterministic and stochastic domains
- 4) Design Kalman Filter and Extended Kalman Filter (widely used signal processing tool)
- 5) Design nonlinear state feedback controllers using backstepping algorithm
- 6) Simplify higher order LTI models to lower order ones

3. Prescribed Text/ Reference Books

1. Control Systems by I.K. Nagrath and M. Gopal, New Age
2. Control Systems Engineering by Norman Nise, John Wiley
3. Modern Control Theory by M. Gopal, New Age
4. Systems and Control by S.H. Zak, OUP
5. Modern Control Design with Matlab and Simulink by Ashish Tewari, John Wiley



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4. Course Plan:

Module	Lecture Sessions
1. Linear Systems Modelling and Analysis (13L, 30%)	L1: Overview of the course and how it takes the previous course forward
	L2-3: Review of state space modelling; State space and transfer function inter-conversion
	L4-6: Similarity transformation; Controllable, observable, diagonal and Jordan canonical forms (graphical and analytical methods)
	L7-8: Solution of state equation; Computation of state transition matrix by Laplace transform and Cayley-Hamilton methods
	L9: Discretization of continuous time state space model by step and impulse invariance methods
	L10-11: BIBO versus asymptotic stability; Controllability, observability, stabilizability, detectability
	L12-13: Introduction to model order reduction techniques: state space/modal/balanced truncation/residualization
2. Nonlinear Systems Modelling and Analysis (9L, 20%)	L14-15: Introduction to nonlinear systems: representations, examples; Hard nonlinearities
	L16-17: Jacobian linearization; Peculiarities of nonlinear systems; Introduction to bifurcation and chaos
	L18-19: Analysis of nonlinear systems: describing function method
	L20: Analysis of nonlinear systems: phase plane method
	L21-22: Stability analysis of nonlinear systems: Lyapunov's indirect and direct methods and LaSalle's invariance principle



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3. Linear and Nonlinear Control Synthesis - I (10L, 25%)	L23-24: Linear state feedback control and its properties; State feedback with integral action
	L25-26: Nonlinear state feedback control: sliding mode control algorithm
	L27-29: Basics of calculus of variations
	L30-32: Optimal linear state feedback: Linear Quadratic Regulator (LQR) and its properties
4. Linear and Nonlinear Control Synthesis - II (10L, 25%)	L33-34: Full order state observer; Disturbance observer
	L35: Observer based state feedback control; Separation principle
	L36: Nonlinear state observer; sliding mode observer
	L37: Stochastic processes; Stationarity and ergodicity; White noise
	L38-40: Stochastic state estimation – discrete time Kalman filter
	L41: Kalman-Bucy filter; Linear Quadratic Gaussian (LQG) control
	L42: Nonlinear stochastic state observer -Extended Kalman Filter; Concluding comments on the course

5. Evaluation Scheme:

Evaluation Component	Weightage (Marks)	Date & Time	Remarks
Midsem	25% (1.5 hr)	07/10 AN	CB+OB
Compre	40% (3 hrs)	08/12 FN	CB+OB
Quizzes 1 & 2 (Announced)	20%	TBA	CB
Assignment	15%	TBA	OB

Closed Book Test: No reference material of any kind will be permitted inside the exam hall.



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Open Book Exam: Handwritten notes (no photocopy) and any one book from the recommended list (photocopy of the book is also fine) will be permitted inside the exam hall. No exchange of any material will be allowed.

6. Chamber Consultation Hour: To be announced in the class

7. Notices: All notices regarding the course will be posted on NALANDA

8. Make-up Policy: Make-up will be granted only to exceptional cases with very strong reasons.

Prof. Dheerendra Singh
Instructor In charge
EEE F422



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